AI & SOCIETY COURSE

Prof. Mor Naaman

Part 5

AI'S WORLDWIDE IMPACT

Unit 3: AI and Democracy

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Where We Are in the Course

Part 1 — AI Foundations
1. What is Generative AI: How it Works and What it Can Do
2. Bias and Limitations: Where GenAI Fails

Part 2 — AI and Individuals
3. AI and You: Cognition, Reliance, and Creativity
4. Persuasion by Design: How AI Influences You

Part 3 — AI, Communication, and Information
5. AI-Mediated Communication and Relationships
6. AI, Culture, and the Information Environment

Part 4 — Challenges for AI Implementation
7. Trust and Safety: AI Harms and Prevention
8. Data, Privacy, and IP: Who Owns What
9. The Challenging Economics of AI

Part 5 — AI's World-wide Impact
10. AI and the World: Education, Work, Power, Democracy
11. Externalities: Human and Environmental Costs of AI

Part 6 — The Way Forward with AI
12. The Future: Navigating AI in Your Community

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This unit: goals

- Understand how AI threatens democracy at three levels: what citizens know; how elections run, and who has power.
- See the opportunities for AI to strengthen democracy

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What democracy requires

> *For a vote to be free and fair, a census must **count the population** and a registration system must ensure that **all (and only) eligible votes** are counted. Citizens should **understand political options**, be able to **form reasonably informed and relevant opinions**, and **organize and express themselves** accordingly. Collective action also entails being able to **exchange ideas with like-minded compatriots**, with **minimal external surveillance**.*

— Danielle Allen & E. Glen Weyl, "The Real Dangers of Generative AI," Journal of Democracy (2024)

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This unit: three risk dimensions

1. **Epistemic risk**
   What citizens can know and believe: bias, persuasion, polarization, and a polluted information environment
2. **Material risk**
   The machinery of democracy: campaigns, election officials, public participation, the vote itself, and surveillance
3. **Foundational risk**
   Who holds power: the capital and compute behind these models concentrate political and economic control

Framing from Summerfield et al. (2025), "The impact of advanced AI systems on democracy"

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This unit: three risk dimensions + opportunity?

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X. **Opportunity?**
   Can AI maybe help democracy?

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1

Epistemic risk: knowledge and information

*** epistemic** *(adj.)*
of or relating to knowledge or knowing

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Why this is a democracy problem

- Democracy assumes citizens can form reasonably informed opinions based on a shared reality and facts.

Four threats:

- Political bias in ordinary, everyday AI use
- AI Persuasion via messages, dialogue, and industrial-scale campaigns
- Polarization through personalization, sycophancy, and reduced social interactions
- The collapse of the information ecosystem

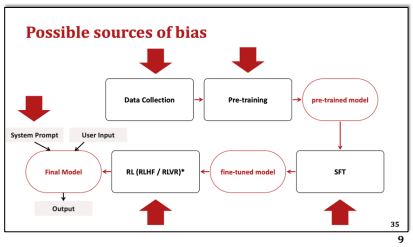
Already covered most of these in Units 1 and 3!

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Political bias, in ordinary use

- Models are not neutral: training data, fine-tuning and system prompts all encode political choices
- Remember from Unit 1



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Political bias (remember this?)

Exercise

Who knows what happened in Tiananmen Square, China in 1989?

Ask Claude, Gemini, or ChatGPT

Now ask DeepSeek, Qwen or Kimi



Source: The Associated Press, originally photographed by Jeff Widener

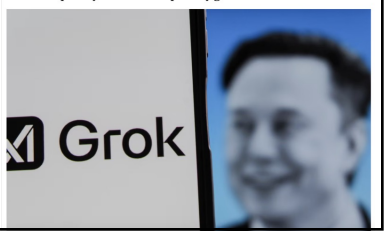
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Political bias (system prompt?)

Elon Musk's Grok AI tells users he is fitter than LeBron James and smarter than Leonardo da Vinci

Users noted that in a raft of now-deleted posts, the chatbot would frequently rank Musk top in any given field



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Political bias (system prompt?)

Latest Newsletters *The Atlantic*

TECHNOLOGY

The Day Grok Told Everyone About 'White Genocide'

What in the world just happened with Elon Musk's chatbot?

By Ali Breland and Matteo Wong



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The models have politics (remember this?)

Persuasion Happens in Natural Use, Too

- Experiment: 1,000 participants interacted with LLMs (Claude 3, Llama 3, and GPT-4) over five exchanges about voting for Trump vs Biden
- The models were NOT prompted to persuade users to support Biden!
- **But about 20% of Trump supporters reduced their support for Trump after LLM interactions**
- Maybe because they were exposed to the actual facts…

Source: Potter et al. (2024), Hidden Persuaders: LLMs' Political Leaning and Their Influence on Voters 26

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The models have politics

- Study in 2023: model opinions are substantially misaligned with US public opinion
- The views of some groups are barely represented at all

Sources: Santurkar et al. (2023), "Whose Opinions Do Language Models Reflect?"

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Political bias, in ordinary use: summary

- Models are not neutral: training data, fine-tuning and system prompts all encode political choices
- The owner can put a thumb on the scale, e.g. Grok, Chinese models
- How the models respond can impact what millions of people think/know!

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Epistemic threats

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Risk: political persuasion via messages, dialogue, and campaigns

Remember? (Unit 2)

Not only conspiracies

Same researchers, new question: can a chatbot move real voters?

Experiments with 1,000s of voters in the elections (US 2024, Canada and Poland 2025). Participants randomly assigned to chat with an AI advocating for one of the top two candidates. AI chat instructed to increase support for the assigned candidate

Shifted opposition voters' candidate ratings by up to

10 points

in Canada and Poland (less in US)

Source: Lin et al. (2025), "Persuading voters using human–AI dialogues" 16

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Risk: political persuasion via messages, dialogue, and campaigns

We saw in unit two:

- **Persuasion via messages/ads by AI** (scale, speed)
- **Persuasion via chat/dialogue** (effective, lasting impact)
  - **Effectiveness relies of fact, but not always accurate…**

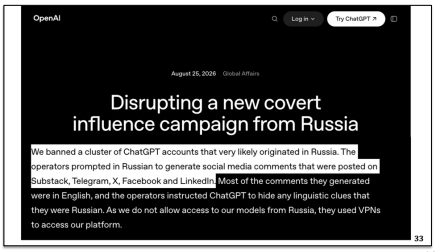
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Risk: political persuasion via messages, dialogue, and campaigns

We saw in Unit 3:

- **Persuasion via information campaigns**

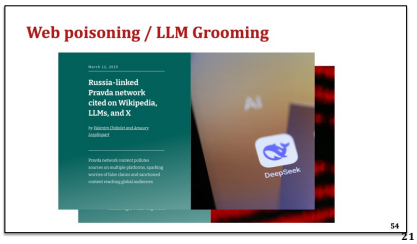


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Also: persuading the model (Unit 3)

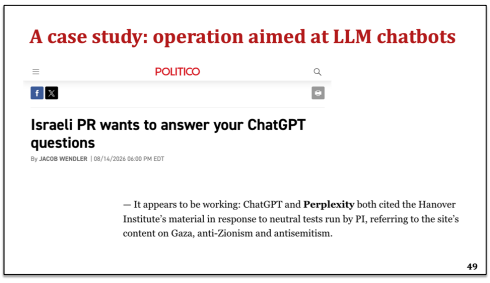
- **"LLM grooming":** create web pages and "reports" on the open web so the models are influenced to use your opinions



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Also: persuading the model (Unit 3)



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Polarization via a model that agrees with you

Polarization *(n.)*

a state in which the opinions, beliefs, or interests of a group or society no longer range along a continuum but become concentrated at opposing extremes

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Polarization via a model that agrees with you

From Unit 2:

Sycophancy Effects

Experiments show that conversations with a sycophantic chatbot increased people:

- ▲ extreme attitudes
- ▲ overconfidence
- ▲ perception that they are "better than average"

Also: people viewed sycophantic chatbots as unbiased, but viewed disagreeable chatbots as highly biased

Source: Rathje, Yu, Globig et al., "Sycophantic AI increases attitude extremity and overconfidence"

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Polarization via a model that agrees with you

- The models are sycophantic (remember?) and personalized: they want to agree with you, provide information that aligns with your history
- It may agree with you politically **and** make you more convicted in your beliefs.
- It may also reduce your motivation to interact with others
- Potential result: **increased polarization**

Source: Kirk, Vidgen, Röttger & Hale (2024), "The benefits, risks and bounds of personalizing the alignment of large language models to individuals"

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Democracy needs a healthy information ecosystem

As we have seen in Unit 3…

- **Deepfakes:** cheap, good enough, and now a routine part of campaigning
- **Degradation of trust:** the bigger harm is not that people believe the fake — it is that they stop believing anything real
- **Decline of reliable sources:** AI answers absorb the traffic and the attention that funded journalism (Unit 3-2)
- **Sheer volume:** synthetic content is becoming a large share of what gets published — and it becomes the training data for the next model

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AI is taking over the information landscape

- Schroeder et al. (2025) call it **malicious AI swarms**: agentic AI plus LLMs, operating as a coordinated network
- Agents that **coordinate autonomously, infiltrate communities, sustain conversations, and fabricate consensus** — cheaply
- AI-generated persuasive content is already rated as **more human-like than content written by humans**
- The same chain-of-thought reasoning that makes models more accurate also makes their falsehoods more convincing

Source: Schroeder et al. (2025), "How malicious AI swarms can threaten democracy," arXiv:2506.06299

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And it is very cheap

$400

a month in compute, for a fully autonomous disinformation system

- A security researcher built **CounterCloud** as a demonstration — in about two months, alone
- It monitored news, generated counter-articles, invented journalist personas, wrote comments and posted them — with no human in the loop
- About 90% of the pipeline was off-the-shelf, open models and standard tooling

Source: Wired (2023), "It costs just $400 to build an AI disinformation machine"

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"The point of modern propaganda isn't only to misinform or push an agenda. It is to exhaust your critical thinking, to annihilate truth."

— Garry Kasparov

(Credit: Bullshit Machines)

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This unit: three risk dimensions + opportunity?

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X **Opportunity?**
Can AI maybe help democracy?

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Material risks: elections, participation, and votes

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From what people believe to what actually happens

- Epistemic risk is about information and knowledge.
- Material risks are about the machinery of democracy: campaigns, officials, public participation, voting process

Four things AI helps:

- Attacking campaigns and election officials — hack-and-leak, spear-phishing
- Flooding public participation until nobody can hear the public
- Suppressing individual voters, one at a time
- Surveilling the opposition at national scale

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Electoral disruption: spear-phishing at scale

Spear-phishing *(n.)*
a targeted cyberattack where criminals send personalized, convincing messages to a specific individual, organization, or department to steal sensitive data or install malware

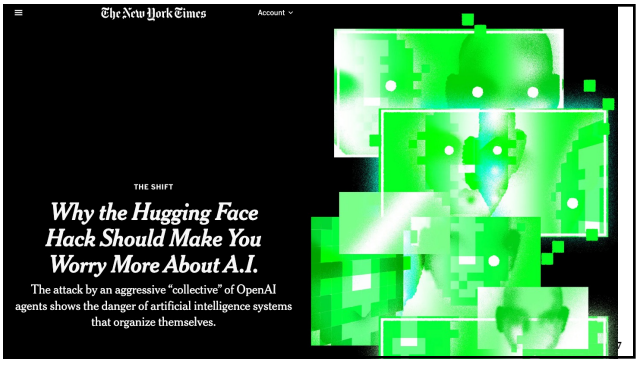
Click-through rates in AI agent experiment:

12%	54%	54%
generic phishing (control)	written by human experts	fully automated AI campaign

Source: Heiding et al. (2024), "Evaluating LLMs' capability to launch fully automated spear phishing campaigns"

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Hurting democratic participation

- Public comment is one of the few places an ordinary person is formally heard
- But AI can now participate at scale…

"As a mayor or a senator, how do I know what really matters to the public anymore?" — State Senator Christopher Cabaldon



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Representatives cannot tell the difference

- If representation is impacted by "what my constituents wrote to me," that signal is now easily overtaken by AI
- Kreps & Kriner sent more than **32,000** constituent emails to about **7,000 US state legislators**; half were machine-written. With **GPT-3, in 2020!**

17.3%
reply rate to emails written by constituents

15.4%
reply rate to emails written by GPT-3

Source: Kreps & Kriner (2023), "The potential impact of emerging technologies on democratic representation"

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Voter suppression via AI

- Vote suppression: discouraging or preventing some people from voting.
- The goal is not to change your vote, but to stop you from voting
- AI can…
  - Create mass targeted voter eligibility challenges
  - Targeted messages with the wrong date, the wrong polling place, or invented ID requirements
  - Personalized harassment of the officials who administer elections
  - Voice cloning (example: January 2024 robocall impersonating a U.S. presidential candidate to tell voters to stay home)

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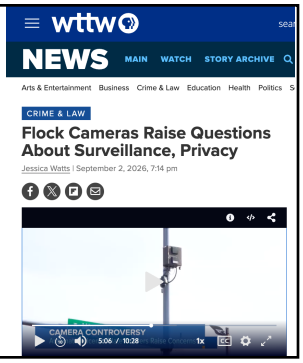
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Surveillance that can target the opposition

"…being able to exchange ideas with like-minded compatriots, with minimal external surveillance." — Allen & Weyl



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Surveillance that can target the opposition

"...being able to exchange ideas with like-minded compatriots, with minimal external surveillance." — Allen & Weyl

The democratic problem: organizing, protesting and dissenting stop being anonymous; opposition can be targeted and intimidated with AI support.

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X **Opportunity?**
Can AI maybe help democracy?

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Foundational risk: who holds the power

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AI companies' power: whose side are they on?

Allen & Weyl's risk framing: a single, coherent will comes to organize and dominate global social life

- These AI firms in a race that rewards speed and scale
- To achieve scale and speed they need resources and compute, so they need political power
- Their valuation/capital and their needs means they are powerful political actors
- Result: these firms have an enormous stake in politics via lobbying, campaign spending, and direct access to government – in the U.S. and beyond.

Source: Allen & Weyl (2024), "The Real Dangers of Generative AI"

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Labor market disruption is a democratic problem

- Remember Units 4 and 5: automation arriving fast; AI's goal is to replace labor!
- Workers are already losing ground to capital – more and more inequality
- The democratic consequence: **inequality, dependency on a ruling class, and the unrest that follows**

Source: Allen & Weyl (2024), "The Real Dangers of Generative AI"

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What if it all happens at once?

Collapse → chaos

- Identification, authentication and the marketplace of ideas stop working
- You cannot run an election on infrastructure nobody trusts
- Waves of automated manipulation, and no shared way to settle what is true

Singularity → concentrated power

- A single coherent will organizes public life: a corporate oligarchy, a technical caste, an authoritarian state, or a very smart AI
- Everyday people will become disempowered

Democratic societies need to defend from all of these.
Authoritarian societies only need to avoid chaos.

Source: Allen & Weyl (2024), "The Real Dangers of Generative AI"

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  Can AI maybe help democracy?

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To be fair: Can AI help democracy?

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Opportunity 1: AI for finding common ground

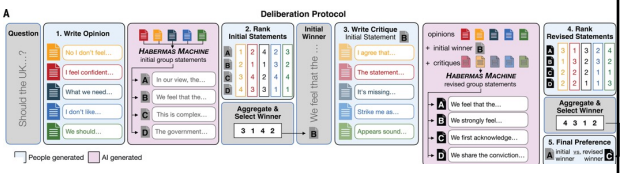
- Tessler et al. (Science, 2024): an LLM (the "Habermas Machine") writes a group statement from five people's individual opinions
- 5,734 UK participants, in groups of five, discuss deep political questions
- The experiment: everyone writes an opinion → the AI drafts candidate group statements → people rank them and write critiques → the AI revises

Slide credit: David Rand.
Source: Tessler et al. (2024), "AI can help humans find common ground in democratic deliberation"

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Finding common ground



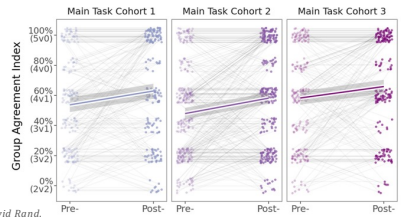
Slide credit: David Rand.
Source: Tessler et al. (2024), "AI can help humans find common ground in democratic deliberation"

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Finding common ground: it works

- Participants preferred the AI-written group statement to one written by a human mediator
- Group agreement rose after deliberation, in every cohort; minority Minority views were represented



Slide credit: David Rand.
Source: Tessler et al. (2024), "AI can help humans find common ground in democratic deliberation"

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Opportunity 2: talking people out of conspiracy theories, other problematic attitudes

From Unit 2:

AI Chat Persuaders (DebunkBot Research)

Chatbots can shift people's conspiracy beliefs – thought to be almost impossible!

2,190 people in an online experiment, each provided a conspiracy they believe in. Then had a chat with GPT-4 Turbo, prompted to persuade the user that their conspiracy theory is unsubstantiated.

Reduced participants belief in their conspiracy

20%
On average

The effect persisted for at least two months

Source: Costello et al. (2024), "Durably reducing conspiracy beliefs through dialogues with AI"

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Opportunity 2: talking people out of conspiracy theories, other problematic attitudes

But remember… it can persuade the other way, too.

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Source: Costello et al. (2024), "Durably reducing conspiracy beliefs through dialogues with AI"

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Opportunity 3: informing people better

- In the persuasion studies, people were swayed by **factual content, delivered patiently and specifically to that person**
- If the model is accurate and trustworthy, this is a net benefit for democracy!
- But it is? And **who decides what "accurate" means?**

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This unit: summary

Many risks posed to democracy from AI.

- **Epistemic:** bias in ordinary use, cheap persuasion at scale, polarization, and an information environment that nobody can trust
- **Material:** easy to target campaigns officers, and officials; public participation is not reliable, voters can be suppressed, and surveillance can help target opposition
- **Foundational:** the capital and compute needed to build these systems concentrate political and economic power in very few hands
- But AI can also help: finding common ground, correcting false beliefs, informing people — if it is accurate, and if we trust whoever decides what accurate means

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